

Assignment -8

Question 1:- Write a function template to swap two variables of any data type.

```
1 //Assignment- 8
2 //Question -1
3 #include <iostream>
4 using namespace std;
5
6 template <class T>
7 void swapValues(T &a, T &b) {
8     T temp = a;
9     a = b;
10    b = temp;
11 }
12
13 int main() {
14     int x = 10, y = 20;
15     swapValues(x, y);
16     cout << "x=" << x << " y=" << y << endl;
17 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  v
x=20 y=10
-----
Process exited after 0.2788 seconds with return value 0
Press any key to continue . . .
```

2. Write a function template to find the minimum element in an array.

```
ques1.cpp  ques2.cpp
1 //Assignment- 8
2 //Question -2
3 #include <iostream>
4 using namespace std;
5
6 template <class T>
7 T findMin(T arr[], int n) {
8     T min = arr[0];
9     for(int i = 1; i < n; i++) {
10         if(arr[i] < min)
11             min = arr[i];
12     }
13     return min;
14 }
15
16 int main() {
17     int arr[] = {5, 2, 9, 1};
18     cout << "Min: " << findMin(arr, 4);
19 }
```

Output :-

```
Min: 1
-----
Process exited after 0.2732 seconds with return value 0
Press any key to continue . . . |
```

3. Write a function template to sort an array using bubble sort.

Code:-

```
ques1.cpp  ques2.cpp  ques3.cpp
1  //Assignment- 8
2  //Question -3
3  #include <iostream>
4  using namespace std;
5
6  template <class T>
7  void bubbleSort(T arr[], int n) {
8      for(int i = 0; i < n-1; i++) {
9          for(int j = 0; j < n-i-1; j++) {
10             if(arr[j] > arr[j+1]) {
11                 T temp = arr[j];
12                 arr[j] = arr[j+1];
13                 arr[j+1] = temp;
14             }
15         }
16     }
17 }
18
19 int main() {
20     int arr[] = {4, 2, 7, 1};
21     bubbleSort(arr, 4);
22
23     for(int i = 0; i < 4; i++)
24         cout << arr[i] << " ";
25 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  ▾
1 2 4 7
-----
Process exited after 0.2323 seconds with return value 0
Press any key to continue . . .
```

4. Write a function template to perform linear search.

Code:-

```
ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp
1  //Assignment- 8
2  //Question -4
3  #include <iostream>
4  using namespace std;
5
6  template <class T>
7  int linearSearch(T arr[], int n, T key) {
8      for(int i = 0; i < n; i++) {
9          if(arr[i] == key)
10             return i;
11     }
12     return -1;
13 }
14
15 int main() {
16     int arr[] = {10, 20, 30};
17     cout << "Index: " << linearSearch(arr, 3, 20);
18 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  x  +  v
Index: 1
-----
Process exited after 0.2522 seconds with return value 0
Press any key to continue . . .
```

5. Design overloaded versions of a function template process() such that:

- i) Accepts a single parameter
- ii) Accepts two parameters of the same type
- iii) Accepts two parameters of different types

Code :-

```
1 //Assignment- 8
2 //Question -5
3 #include <iostream>
4 using namespace std;
5
6 // i) Single parameter
7 template <class T>
8 void process(T a) {
9     cout << "One parameter: " << a << endl;
10 }
11
12 // ii) Same type
13 template <class T>
14 void process(T a, T b) {
15     cout << "Same type: " << a << ", " << b << endl;
16 }
17
18 // iii) Different types
19 template <class T, class U>
20 void process(T a, U b) {
21     cout << "Different types: " << a << ", " << b << endl;
22 }
23
24 int main() {
25     process(10);
26     process(10, 20);
27     process(10, 3.5);
28 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  x + v
One parameter: 10
Same type: 10, 20
Different types: 10, 3.5

-----
Process exited after 0.2471 seconds with return value 0
Press any key to continue . . .
```

6. Develop a class template for stack with push and pop operations.

Code:-

ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp [*] Untitled6

```
1 //Assignment- 8
2 //Question -6
3 #include <iostream>
4 using namespace std;
5 template <class T>
6 class Stack {
7     T arr[100];
8     int top;
9 public:
10     Stack() { top = -1; }
11
12     void push(T x) {
13         arr[++top] = x;
14     }
15
16     void pop() {
17         if(top >= 0)
18             top--;
19     }
20
21     void display() {
22         for(int i = top; i >= 0; i--)
23             cout << arr[i] << " ";
24     }
25 };
26 int main() {
27     Stack<int> s;
28     s.push(10);
29     s.push(20);
30     s.display();
31 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ x + v
20 10
-----
Process exited after 0.3011 seconds with return value 0
Press any key to continue . . .
```

7. Develop a class template for queue with enqueue and dequeue operations.

Code:-

```

ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp [*] Untitled7
1 //Assignment- 8
2 //Question -7
3 #include <iostream>
4 using namespace std;
5 template <class T>
6 class Queue {
7     T arr[100];
8     int front, rear;
9 public:
10    Queue() {
11        front = 0;
12        rear = -1;
13    }
14    void enqueue(T x) {
15        arr[++rear] = x;
16    }
17    void dequeue() {
18        if(front <= rear)
19            front++;
20    }
21    void display() {
22        for(int i = front; i <= rear; i++)
23            cout << arr[i] << " ";
24    }
25 };
26
27 int main() {
28     Queue<int> q;
29     q.enqueue(5);
30     q.enqueue(10);
31     q.display();
32 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  X + v
5 10
-----
Process exited after 0.3108 seconds with return value 0
Press any key to continue . . .

```

8. Create a class template to store and display a pair of values.

Code:-

```

ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp ques7.cpp ques8.cpp
1 //Assignment- 8
2 //Question -8
3 #include <iostream>
4 using namespace std;
5 template <class T, class U>
6 class Pair {
7     T a;
8     U b;
9 public:
10    Pair(T x, U y) {
11        a = x;
12        b = y;
13    }
14
15    void display() {
16        cout << a << " , " << b;
17    }
18 };
19
20 int main() {
21     Pair<int, float> p(10, 5.5);
22     p.display();
23 }

```

Output:-

```
C:\Users\asus\Desktop\oops\  X + v
10 , 5.5
-----
Process exited after 0.2779 seconds with return value 0
Press any key to continue . . .
```

9. Create a class template for basic arithmetic operations.

Code:-

```
ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp  ques5.cpp  ques6.cpp  ques7.cpp  ques8.cpp  ques9.cpp
1  //Assignment- 8
2  //Question -9
3  #include <iostream>
4  using namespace std;
5  template <class T>
6  class Arithmetic {
7      T a, b;
8  public:
9      Arithmetic(T x, T y) {
10         a = x;
11         b = y;
12     }
13
14     void calculate() {
15         cout << "Sum: " << a + b << endl;
16         cout << "Diff: " << a - b << endl;
17         cout << "Product: " << a * b << endl;
18         cout << "Division: " << a / b << endl;
19     }
20 };
21
22 int main() {
23     Arithmetic<int> obj(10, 5);
24     obj.calculate();
25 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  X + v
Sum: 15
Diff: 5
Product: 50
Division: 2
-----
Process exited after 0.2798 seconds with return value 0
Press any key to continue . . .
```

10. Create a class template to input and display array elements.

Code:-

```
ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp ques7.cpp ques8.cpp ques9.cpp ques10.cpp
1 //Assignment- 8
2 //Question-10
3 #include <iostream>
4 using namespace std;
5 template <class T>
6 class Array {
7     T arr[100];
8     int n;
9
10 public:
11     void input(int size) {
12         n = size;
13         for(int i = 0; i < n; i++)
14             cin >> arr[i];
15     }
16
17     void display() {
18         for(int i = 0; i < n; i++)
19             cout << arr[i] << " ";
20     }
21 };
22
23 int main() {
24     Array<int> a;
25     a.input(3);
26     a.display();
27 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ x + v
8 7 6
8 7 6
-----
Process exited after 2.632 seconds with return value 0
Press any key to continue . . .
```